

Explicit Results for the Distributions of Queue Lengths for a Non-Preemptive Two-Level Priority Queue

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Explicit results are derived using simple and exact methods for the joint and marginal queue-length distributions for the M/M/c queue with two non-preemptive priority levels. Equal service rates are assumed. Two approaches are considered. One is based on numerically robust quadratic recurrence relations. The other is based on a complex contour-integral representation that yields exact closed-form analytical expressions, not hitherto available in the literature, that can also be evaluated numerically with very high accuracy.

Key words: queueing theory; non-preemptive priority; queue length distribution

MSC2000 subject classification: Primary: 90B22; secondary: 60K25, 60J74

OR/MS subject classification: Primary: Queues: Priority; secondary: Queues: Markovian

History: Date created: July 03, 2023. Last update: September 14, 2023.

1. Introduction This work is concerned with the development of practical algorithms for the computation of joint and marginal distributions of queue lengths for the M/M/c queue with a non-preemptive priority discipline. Applications of this model are found in telecommunications [5], health care [10, 25], radar [21, 22], air traffic control [23] and numerous other areas.

The **non-preemptive** priority queue discipline is as stated by Dressin and Reich [7]: Once a client's service has begun, it is permitted to proceed to completion. If a server becomes empty, and there is at least one client waiting, then a client of the highest priority present in the queue is admitted to the server. Clients of equal priority are served on a **first-come, first-served** basis. This is also known as the 'head of the line' discipline.

Thus, let us consider a non-preemptive queue with K priority levels, each with a distinct Poisson arrival rate λ_k , $k = 1, 2, \dots, K$ and corresponding level traffic intensity¹ $r_k = \lambda_k / (N\mu)$, leading to a total traffic intensity for the aggregation of all arrivals of $r = \sum_{k=0}^K r_k$. We adopt the usual convention that smaller **priority-level indices k represent higher priorities**. Thus, r_1 denotes the traffic intensity associated with the highest priority level. For simplicity, we have assumed a common exponential service rate μ among all priority levels. The number of servers is denoted by $c = N$.

In this work, attention is confined to the two-level problem $K = 2$. Analysis of this case is amenable to a number of analytical techniques that do not extend easily, or at all, to the general multi-level priority problem. Also, the two-level problem has a distinguished status. All marginal distributions for the multi-level problem can be inferred from the low-priority marginal pertaining to just two priority levels [6]. If we let r_{hi} and r_{lo} denote the level traffic intensities for the high and low priority arrivals, respectively, for the two-level problem, then the wait-conditional²

¹ Consistent with [8, 11], the notation ρ_k reserved for $\rho_k \equiv \lambda_k / \mu$, so that $r_k = \rho_k / N$.

² See below in Section 2.

marginal distribution of the queue length for priority level $k = 1, 2, \dots, K$ in the multi-level problem is obtained by making the identifications

$$r_{\text{lo}} = r_k, \quad r_{\text{hi}} = \sum_{\ell=1}^{k-1} r_{\ell}, \quad (1)$$

so that the total traffic intensity in the effective (wait-conditional) two-level problem becomes $r = r_{\text{sum}}$, with

$$r_{\text{sum}} = r_{\text{lo}} + r_{\text{hi}} = \sum_{\ell=1}^k r_{\ell}. \quad (2)$$

For the actual two-level problem, we have the identifications $r_{\text{hi}} \equiv r_1$, $r_{\text{lo}} \equiv r_2$, and we shall use both sets of notation interchangeably. It is also convenient to introduce the parameter ν that represents the fraction of all arrivals that are of high priority (which we abbreviate as ‘hifrac’). Thus $r_{\text{hi}} = \nu r$, $r_{\text{lo}} = (1 - \nu)r$, $0 \leq \nu \leq 1$.

Previous work on the non-preemptive priority queue has focused, almost entirely, on calculating moments and the waiting-time distributions per priority level. In early work, Cobham [3, 4], followed by Holley [9], were first to consider the mean waiting times and queue lengths. Waiting-time means and second moments for general service-time distributions were subsequently given by Kesten and Runnenberg [15]. Gail et al. [8] studied the non-preemptive M/M/c system for two priority levels with different exponential service rates. While they developed a matrix algorithm for determining various characteristics of a generating function for this problem, explicit results were also confined to the mean waiting times and queue lengths.

For the waiting-time problem, Davis [6] improved on previous work by Dressin and Reich [7] to derive an explicit integral expression for the waiting-time distribution for the non-preemptive priority queue. He analysed the two-level problem, as the waiting time distribution for the multi-level problem can be inferred from the two-level case. He did not study the queue-length marginals, and they cannot be directly inferred from the waiting-time distributions by appealing to the distributional form of Little’s law [1, 13] as the no-overtaking assumption is violated. Kella and Yechiali [14] covered the same ground as Davis for the probability waiting function (PGF) of the waitingtime, but using a different methodology. The moment generating function (MGF) of the waiting time and associated moments have also been considered in [19]. More recently, Wagner [26] has studied the waiting-time MGF for a finite-capacity, multi-server version of the same problem as Davis.

For the queue-length distributions, Miller [17, 18] uses a matrix-geometric method for the two-level problem that results in a complex algorithm involving multiple levels of recursion. Little is presented about the numerical stability of this approach, and it is known to deteriorate for traffic intensities close to unity. Kao and Narayanan [11] and Kao and Wilson [12] also employ matrix-geometric methods for the two-level problem which, as they point out, unavoidably require finite-state truncation. The aforementioned papers deal with unequal service rates. The matrix-geometric method [20] applied to queueing models has the singular disadvantage that it necessitates truncation of the problem to prescribed finite maximum values of queue lengths. While powerful, it is complex and not elegant. Thus, its use should best be avoided whenever simpler alternatives are available, and this is manifestly the case for the present problem, as will become clear.

In earlier work, Marks [16] studied the two-level problem with common service rate and derived a highly complex system of linear partial difference equations that must be solved recursively. The required manipulations are cumbersome and no insight into the analytic structure of the problem is gained. However, it is most likely the first paper where actual queue-length probabilities, rather than the PGF, were computed. No light is shed on the numerical stability of the method.

A different approach, based on a partial PGF, is due to Cohen [5], who studied the two-level problem with equal service rates; and it is this approach that we pursue in the discussion that

follows. We take up the programme where Cohen [5] left off, in devising simple and practical schemes for extracting actual probabilities from the PGF. There is the additional benefit that this approach can be extended to the general multi-level problem. Shortle et al. [24] have remarked that ‘the determination of stationary probabilities in a non-preemptive Markovian system is an exceedingly difficult matter, well near impossible when the number of priorities exceeds two’. In a separate forthcoming paper, we shall demonstrate otherwise.

The present work focuses on explicit results that are useful for practical applications. While we do not purport to have made general theoretical advances in priority queues, the work does serve to fill a large gap in the literature by establishing basic results for a paradigmatic model that one would expect to have been uncovered decades ago. We believe that it also has pedagogical value. For the two-level non-preemptive priority queue, Shortle et al. [24], in the most recent edition of their textbook, set up the stationary balance equations but remark that ‘obtaining a reasonable solution to these stationary equations is very difficult, ... The most we can do comfortably is obtain expected values via two-dimensional generating functions’. The simplicity of the methods described herein might render a more detailed treatment of the subject suitable for elementary texts.

2. Non-Preemptive Priority Queue The no-wait probability P_{NW} is the probability that a new arrival will find at least one server idle. It is clearly independent of the queue discipline, and is given by [6]

$$\frac{1}{1 - P_{\text{NW}}} = 1 + (1 - r) \frac{N!}{(Nr)^N} \cdot \sum_{k=0}^{N-1} \frac{(Nr)^k}{k!}. \quad (3)$$

Let $P(n, m)$ denote the steady-state probability that there are n low-priority clients in the queue (rather than in the system) and m high-priority clients in the queue. We have the decomposition [cf. 6]

$$P(n, m) = P_{\text{NW}} \cdot \delta_{n0} \delta_{m0} + (1 - P_{\text{NW}}) \cdot f(n, m), \quad (4)$$

where $f(n, m)$ represents the wait-conditional joint PMF, *i.e.* the probability that there are n low-priority clients and m high-priority clients in the queue, given that all servers are busy. The wait-conditional distribution does not explicitly depend on the number of servers N . There is only an indirect dependence on N through the total traffic intensity r .

Our starting point is the paper of Cohen [5], which introduced a partial PGF for the problem that summed only over the low-priority argument:

$$G_m(p) \equiv \sum_{n=0}^{\infty} p^n P(n, m). \quad (5)$$

This turns out to be a very convenient strategy, especially given the fact that the wait-conditional high-priority marginal is a simple geometric distribution. Only the low-priority marginal is non-trivial. We introduce a wait-conditional version $g_m(p)$ of this PGF such that

$$g_m(p) \equiv \sum_{n=0}^{\infty} p^n f(n, m). \quad (6)$$

It follows that

$$G_m(p) = P_{\text{NW}} \cdot \delta_{m0} + (1 - P_{\text{NW}}) \cdot g_m(p). \quad (7)$$

Cohen’s result [5] for the wait-conditional PGF for the two-level non-preemptive priority queue with equal service rates is³

$$g_m(p) = \frac{(1 - r)(1 - p)}{1 - p\lambda_2(p)} \cdot \lambda_1^m(p), \quad (8)$$

³ The quantities $\lambda_{1,2}(p)$ should not be confused with the arrival rates introduced earlier. We are adhering to Cohen’s original, but less than ideal, notation.

where $\lambda_{1,2}(p)$ are defined as follows: Let us introduce $\lambda(p) = \lambda_{\pm}(p)$ as the two solutions of the quadratic equation

$$\lambda^2 - (1 + r - r_2 p)\lambda + r_1 = 0, \quad (9)$$

such that

$$\lambda_{\pm}(p) = [b(p) \pm \sqrt{b^2(p) - 4r_1}]/2, \quad b(p) \equiv 1 + r - r_2 p. \quad (10)$$

Then in (8), we have $\lambda_1(p) = \lambda_-(p)$, $\lambda_2(p) = \lambda_+(p)$, and it is useful to note that

$$\lambda_+(p) + \lambda_-(p) = b(p), \quad \lambda_+(p) \cdot \lambda_-(p) = r_1. \quad (11)$$

Another way to express the PGF for the wait-conditional distribution is

$$g_m(p) = g_{lo}(p) \cdot [1 - \lambda_1(p)] \lambda_1^m(p), \quad g_{lo}(p) = \frac{1 - r}{\lambda_2(p) - r}. \quad (12)$$

That $g_{lo}(p)$ represents the wait conditional PGF for the low-priority marginal is clear from observing that

$$\sum_{m=0}^{\infty} g_m(p) = g_{lo}(p). \quad (13)$$

On the other hand, it follows directly from (12) that the wait-conditional PMF for the high priority marginal is given by

$$f_{hi}(m) = g_m(1) = (1 - r_1) r_1^m. \quad (14)$$

Consequently, the only marginal distribution of interest in the present study is that for the low-priority level.

By construction, the wait-conditional joint PMF $f(n, m)$ is recovered from the PGF $g_m(p)$ according to

$$f(n, m) = \frac{1}{n!} \cdot \frac{d^n}{dp^n} g_m(p) \Big|_{p=0}. \quad (15)$$

The multiple derivative is prohibitively cumbersome to directly perform analytically. Thus, we proceed to present two alternative strategies that render the problem tractable.

3. Quadratic Recurrence The first method constructs a recurrence relation based on the fact that the functions $\lambda_{\pm}(p)$ solve a quadratic equation. We begin by considering the low-priority marginal, whose PGF can be expressed as

$$g_{lo}(p) = \frac{1 - r}{\lambda_2(p) - r}. \quad (16)$$

Since $\lambda_2(p)$ satisfies a quadratic equation, then so does $g_{lo}(p)$. Let us set

$$u \equiv r_2 p, \quad g(u) \equiv \frac{1}{\lambda_2 - r} = \sum_{k=0}^{\infty} \frac{g_k}{k!} u^k \quad \Rightarrow \quad g_k = \frac{d^k g(u)}{du^k} \Big|_{u=0}. \quad (17)$$

Then we obtain

$$(ru - r_2)g^2 + (u - 1 + r)g + 1 = 0. \quad (18)$$

We now differentiate this equation n times with respect to u , and use the identities

$$\begin{aligned} \frac{1}{n!} \cdot \frac{d^n}{du^n} (ug) \Big|_{u=0} &= \frac{g_{n-1}}{(n-1)!}, \\ \frac{1}{n!} \cdot \frac{d^n}{du^n} (g^2) \Big|_{u=0} &= \sum_{k=0}^n \frac{g_k}{k!} \cdot \frac{g_{n-k}}{(n-k)!}, \\ \frac{1}{n!} \cdot \frac{d^n}{du^n} (ug^2) \Big|_{u=0} &= \sum_{k=0}^{n-1} \frac{g_k}{k!} \cdot \frac{g_{n-k-1}}{(n-k-1)!}. \end{aligned} \quad (19)$$

For the quantities $f_k \equiv g_k/k!$, this leads to the non-linear recurrence relations

$$f_n = \frac{1 + rf_0}{1 - r + 2r_2f_0} \cdot f_{n-1} + \frac{1}{1 - r + 2r_2f_0} \sum_{k=1}^{n-1} f_k \cdot (rf_{n-k-1} - r_2f_{n-k}), \quad (20)$$

for $n = 1, 2, \dots$, with

$$f_0 = \frac{1}{2r_2} \left[\sqrt{(1-r)^2 + 4r_2} - (1-r) \right] = \frac{2}{1 - r + \sqrt{(1-r)^2 + 4r_2}} > 0. \quad (21)$$

The expression for f_0 follows from $f_0^{-1} = g_0^{-1} = \lambda_2(0) - r$. We observe that $f_{10}(n) = (1-r)r_2^n f_n$.

Efficient vectorized implementations in MATLAB are possible. Practical implementation proceeds as follows: Let us introduce an arbitrary scale factor Λ , define

$$D \equiv 1 - r + 2r_2f_0 = \sqrt{(1-r)^2 + 4r_2}, \quad c_1 \equiv r_2/\Lambda, \quad c_2 \equiv \Lambda/D, \quad (22)$$

and scale according to $\tilde{f}_n \equiv \Lambda^n f_n = (r_2/c_1)^n f_n$. Then we solve the recurrence

$$\tilde{f}_n = c_2 \cdot \left(\tilde{f}_{n-1} + \sum_{k=0}^{n-1} \tilde{f}_k \Delta_{(n-1)-k} \right) \quad (23)$$

and recover the marginal as $f_{10}(n) = (1-r)c_1^n \tilde{f}_n$. At each step, we set

$$\Delta_k \equiv r\tilde{f}_k - c_1\tilde{f}_{k+1}, \quad (24)$$

for $k = 1, 2, \dots, n-1$, subject to the initialization $\tilde{f}_n \leftarrow 0$ within the scope of evaluating Δ_{n-1} . We find that good numerical performance is achieved with $\Lambda = r_2$, so that $c_1 = 1$.

Analogous treatment of the joint PMF is only marginally more complex. Based on the quadratic

$$\lambda_{\pm}^2 + (u-1-r)\lambda_{\pm} + r_1 = 0, \quad (25)$$

we solve for the Taylor-series coefficients $\lambda_{\pm}^{(k)}$ in

$$\lambda_{\pm} = \sum_{k=0}^{\infty} \lambda_{\pm}^{(k)} p^k = \sum_{k=0}^{\infty} \Lambda^{-k} f_k^{\pm} u^k, \quad (26)$$

for some arbitrary scale factor Λ , using the non-linear recurrence

$$f_n^{\pm} = \mp \frac{1}{\sqrt{(1-r)^2 + 4r_2}} \left(\Lambda f_{n-1}^{\pm} + \sum_{k=1}^{n-1} f_k^{\pm} \cdot f_{n-k}^{\pm} \right), \quad (27)$$

$n = 1, 2, \dots$, where

$$f_0^{\pm} = \frac{1}{2} \left(1 + r \pm \sqrt{(1-r)^2 + 4r_2} \right). \quad (28)$$

The λ -coefficients are recovered according to $\lambda_{\pm}^{(k)} = (r_2/\Lambda)^k f_k^{\pm}$. As with the marginal, the choice $\Lambda = r_2$ results in good numerical performance. All that remains to be done is to use the standard recursion for multiplication of power series as dictated by (12). The simplest way to proceed is via repeated convolutions:

$$\begin{aligned} \phi_0 &= (1-r) \cdot \text{conv} \left(\frac{1}{\lambda_2 - r}, 1 - \lambda_1 \right), \\ \phi_k &= \text{conv}(\phi_{k-1}, \lambda_1), \end{aligned} \quad (29)$$

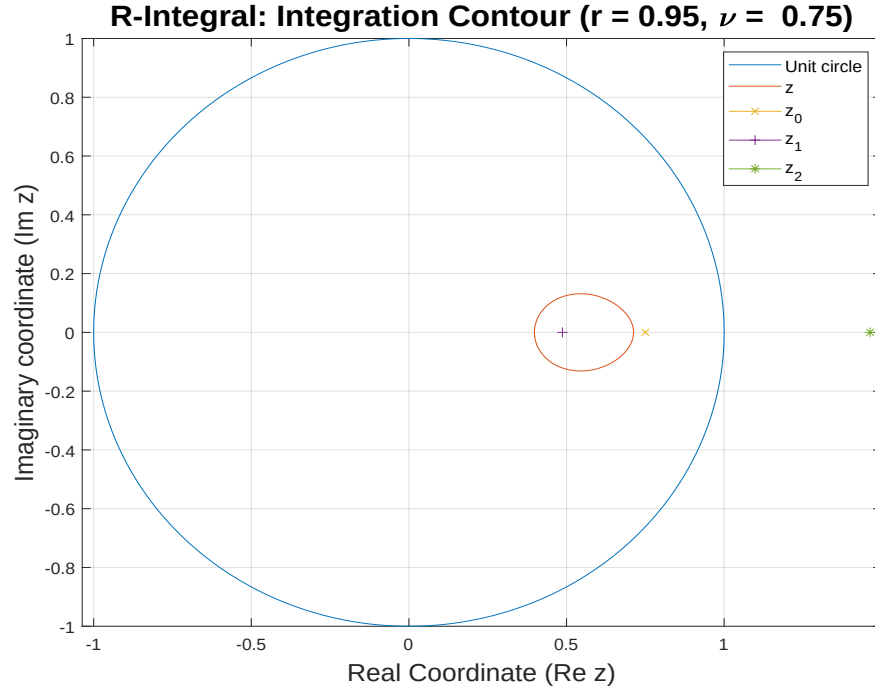


FIGURE 1.

Note. The z -contour that results from taking the p -contour to be the unit circle centred on the origin, plotted for the case of total traffic intensity $r = 0.95$ and fraction of high-priority arrivals $\nu = 0.75$. Also displayed are the locations of R-integral poles z_0, z_1, z_2 .

for $k = 1, 2, \dots, m$. Then $f(n, m) = \phi_m(n)$. The conv function is defined like the MATLAB function of the same name: Suppose that $C(u) = A(u)B(u)$, with

$$A(u) = \sum_{n=0}^{n_1} a(n)u^n, \quad B(u) = \sum_{n=0}^{n_2} b(n)u^n, \quad C(u) = \sum_{n=0}^{n_1+n_2} c(n)u^n. \quad (30)$$

Then $c = \text{conv}(a, b)$, where

$$c(n) = \text{conv}(a, b)(n) \equiv \sum_{k=0}^n a(k)b(n-k), \quad (31)$$

for $n = 0, 1, 2, \dots, n_1 + n_2$.

The foregoing recurrence relations constitute a significant improvement over the strategy implemented in [2], and are vastly simpler than those arising from the matrix-geometric method as considered in [11, 12, 17, 18]. While the quadratic recurrence method exhibits excellent numerical behaviour, it gives little insight into the analytical structure of the distributions. This deficiency is addressed in the next section.

4. Complex Contour Integral Another strategy in dealing with (15) is to represent it in terms of a complex contour integral in accordance with Cauchy's integral theorem. This yields

$$f(n, m) = (1-r) \oint_C \frac{dp}{2\pi i} \frac{(1-p)\lambda_1^m}{p^{n+1}(1-p\lambda_2)}, \quad (32)$$

where $\mathcal{C}$ is an anti-clockwise circle centred about the origin with radius less than $1/r$. It follows directly that the low-priority marginal PMF, defined by

$$f_{\text{lo}}(n) \equiv \sum_{m=0}^{\infty} f(n, m), \quad (33)$$

is represented as a complex contour integral by

$$f_{\text{lo}}(n) = (1-r) \oint_{\mathcal{C}} \frac{dp}{2\pi i} \frac{1-p}{p^{n+1}} \cdot \frac{1}{(1-p\lambda_2)(1-\lambda_1)}. \quad (34)$$

The conventional approach in dealing with such contour integrals, mirroring the approach adopted previously for the waiting-time distribution [6], would be to deform the contour by expanding it to the circle at infinity while avoiding a cut of finite extent on the real axis that is generated by the square-root component of $\lambda_{\pm}(p)$, and a possible simple pole that also lies on the real axis. The circle at infinity yields a vanishing contribution, which leaves a (potential) pole term and a real-valued integral along the cut. We shall explore this approach separately in a forthcoming paper, where we shall show that it leads to integral expressions that are amenable to efficient quadrature algorithms, and can also be evaluated analytically in terms of a generalized form of the associated Legendre functions. In the present work, we pursue a different method based on a change of integration variable.

Let $\lambda = z_1, z_2$ be the roots of the polynomial equation $\lambda^2 - (1+r)\lambda + r_1 = 0$, so that we have $z_1 + z_2 = 1+r$, $z_1 z_2 = r_1 = \nu r$. Then, the inversion of $z = \lambda_{\pm}(p)$ yields

$$p = -(z - z_1)(z - z_2)/(r_2 z). \quad (35)$$

Thus,

$$dp = -\frac{1}{r_2} \left(1 - \frac{z_1 z_2}{z^2}\right) dz, \quad (36)$$

in which case

$$\frac{dp}{p^{n+1}} = (-r_2)^n \frac{(z^2 - z_1 z_2) z^{n-1}}{[(z - z_1)(z - z_2)]^{n+1}} \cdot dz. \quad (37)$$

We make the change of integration variable $p \mapsto z : z = \lambda_1(p)$, in which case $\lambda_2(p) = r/z$, and we make the identifications

$$z_0 = r_1/r, \quad z_1 = \lambda_-(p=0), \quad z_2 = \lambda_+(p=0), \quad (38)$$

or, equivalently,

$$z_0 = \nu, \quad z_1 = \frac{1}{2} \left[1 + r - \sqrt{(1+r)^2 - 4\nu r}\right], \quad z_2 = \frac{1}{2} \left[1 + r + \sqrt{(1+r)^2 - 4\nu r}\right]. \quad (39)$$

Then, we obtain

$$\frac{1-p}{1-p\lambda_2} = \frac{z}{r} \cdot \frac{z-1}{z-z_0}. \quad (40)$$

It follows that the joint PMF is given by

$$f(n, m) = \frac{(1-r)(-r_2)^n}{r} \oint_{\mathcal{C}'} \frac{dz}{2\pi i} \frac{z^{m+n}}{z-z_0} \cdot \frac{(z-1)(z^2 - z_1 z_2)}{[(z - z_1)(z - z_2)]^{n+1}}, \quad (41)$$

where $\mathcal{C}'$ is a closed anti-clockwise contour that encloses the pole at $z = z_1$. but with the poles at $z = z_0, z_2$ in the exterior. For the low-priority marginal PMF, we have

$$f_{\text{lo}}(n) = -\frac{(1-r)(-r_2)^n}{r} \oint_{\mathcal{C}'} \frac{dz}{2\pi i} \frac{z^n}{z-z_0} \cdot \frac{(z^2 - z_1 z_2)}{[(z - z_1)(z - z_2)]^{n+1}}. \quad (42)$$

5. R-Integrals In order to evaluate the integral representations for the joint and marginal PMFs, derived the foregoing section, we introduce a collection of complex contour integrals, to which we shall refer as the R-integrals, according to the definition

$$R_n^m \equiv \oint_{\mathcal{C}'} \frac{dz}{2\pi i} \frac{1}{z - z_0} \cdot \frac{z^m}{[(z - z_1)(z - z_2)]^n}, \quad (43)$$

for $m, n = 0, 1, 2, \dots$, where $\mathcal{C}'$ is a closed anti-clockwise contour that encloses the pole at $z = z_1$, but with the poles at $z = z_0, z_2$ in the exterior. An immediate consequence of this definition is the (backwards) recurrence relation

$$R_{n-1}^m = R_n^{m+2} - (z_1 + z_2)R_n^{m+1} + z_1 z_2 R_n^m. \quad (44)$$

One may also note the scaling behaviour

$$R_n^m(z_0, z_1, z_2) = z_0^{m-2n} R_n^m(1, z_1/z_0, z_2/z_0), \quad (45)$$

or, more generally,

$$R_n^m(z_0, z_1, z_2) = \zeta^{m-2n} R_n^m(z_0/\zeta, z_1/\zeta, z_2/\zeta), \quad (46)$$

for any $\zeta > 0$.

In the present application to the priority queue, the parameters z_0, z_1, z_2 are given by (39). In terms of the R-integrals, the joint PMF is given by

$$f(n, m) = \frac{(1-r)(-r_2)^n}{r} (R_{n+1}^{m+n+3} - R_{n+1}^{m+n+2} - z_1 z_2 R_{n+1}^{m+n+1} + z_1 z_2 R_{n+1}^{m+n}). \quad (47)$$

If we introduce the difference functions $\Delta R_n^m \equiv R_n^{m+1} - R_n^m$, then we can write

$$f(n, m) = \frac{(1-r)(-r_2)^n}{r} (\Delta R_{n+1}^{m+n+2} - z_1 z_2 \Delta R_{n+1}^{m+n}). \quad (48)$$

Likewise, in terms of the R-integrals, we have for the low-priority marginal PMF,

$$f_{\text{lo}}(n) = -\frac{(1-r)(-r_2)^n}{r} (R_{n+1}^{n+2} - z_1 z_2 R_{n+1}^n), \quad (49)$$

for $n = 0, 1, 2, \dots$. For the exclusively-low distribution, defined by $f_{\text{xlo}}(n) \equiv f(n, 0)$, we can write

$$f_{\text{xlo}}(n) = \frac{(1-r)(-r_2)^n}{r} (\Delta R_{n+1}^{n+2} - z_1 z_2 \Delta R_{n+1}^n). \quad (50)$$

It gives the probability of finding n low-priority clients in the queue and no high-priority clients. It has a form that is similar to the low-priority marginal $f_{\text{lo}}(n)$, and we will show later that the two are, in fact, closely related. This relationship will provide a useful diagnostic test of the numerical performance of the R-integral computation.

We have succeeded in recasting the problem into one that involves complex contour integration over a collection of totally meromorphic functions. In Figure 1, we plot the z -contour $\mathcal{C}'$ that results from taking the p -contour $\mathcal{C}$ to be the unit circle centred on the origin, plotted for the case of total traffic intensity $r = 0.95$ and fraction of high-priority arrivals $\nu = 0.75$. Also displayed are the locations of R-integral poles z_0, z_1, z_2 .

5.1. Recurrence If we cast the recurrence relation (44) as

$$R_n^m = R_{n-1}^{m-2} - z_1 z_2 R_n^{m-2} + (z_1 + z_2) R_n^{m-1}, \quad (51)$$

for $m = 2, 3, \dots$, $n = 1, 2, \dots$, then it may, in principle, be solved recursively for the R_n^m starting from the seed values

$$\begin{aligned} R_0^m &= 0, \\ R_{n+1}^0 &= \frac{(-1)^n}{[(z_1 - z_0)(z_1 - z_2)]^{n+1}} \cdot p_n \left(\frac{z_1 - z_0}{z_1 - z_2} \right), \\ R_{n+1}^1 &= \frac{(-1)^n}{(z_1 - z_2)^{2n+1}} \binom{2n}{n} + z_0 R_{n+1}^0, \end{aligned} \quad (52)$$

with the polynomials $p_n(x)$ defined by

$$p_n(x) \equiv \sum_{k=0}^n \binom{k+n}{k} x^k. \quad (53)$$

Unfortunately, this recursion scheme is numerically unstable, especially for small ν .

5.2. Series Representation Applying Cauchy's theorem to (43), followed by an invocation of Leibniz's formula, we obtain

$$\begin{aligned} R_{n+1}^m &= \frac{1}{n!} \cdot \frac{d^n}{dz_1^n} \left[\frac{1}{(z_1 - z_2)^{n+1}} \cdot \frac{z_1^m}{z_1 - z_0} \right] \\ &= \sum_{k=0}^n \frac{1}{(n-k)!} \frac{d^{n-k}}{dz_1^{n-k}} \left[\frac{1}{(z_1 - z_2)^{n+1}} \right] \cdot \frac{1}{k!} \frac{d^k}{dz_1^k} \left[\frac{z_1^m}{z_1 - z_0} \right]. \end{aligned} \quad (54)$$

The first differentiation is trivial to perform, yielding

$$R_{n+1}^m = (-1)^n \sum_{k=0}^n \binom{2n-k}{n} \frac{z_0^{m-k-1}}{(z_2 - z_1)^{2n+1-k}} S_k^m(z_1/z_0), \quad (55)$$

where

$$S_k^m(x) \equiv \frac{1}{k!} \frac{d^k}{dx^k} \left(\frac{x^m}{1-x} \right). \quad (56)$$

The functions $S_k^m(x)$ satisfy the relationship

$$S_k^{m+1}(x) - S_k^m(x) = -\binom{m}{k} x^{m-k}. \quad (57)$$

It is convenient to introduce polynomials

$$P_k^m(x) \equiv (1-x)^{k+1} S_k^m(x), \quad (58)$$

so that $P_k^0(x) = 1$, $P_0^m(x) = x^m$. Then we can write

$$R_{n+1}^m = \frac{(-1)^n}{(z_2 - z_1)^{2n+2}} \sum_{k=0}^n \binom{2n-k}{n} \left(\frac{z_2 - z_1}{1 - z_1/z_0} \right)^{k+1} z_0^{m-k-1} P_k^m(z_1/z_0). \quad (59)$$

Combining (56) and (58), we can establish that, for $m > k$,

$$\begin{aligned} P_k^m(x) &= (1-x)^{k+1} \sum_{\ell=0}^k \frac{1}{\ell!} \frac{d^\ell x^m}{dx^\ell} \cdot \frac{1}{(k-\ell)!} \frac{d^{k-\ell}}{dx^{k-\ell}} \left(\frac{1}{1-x} \right) \\ &= \sum_{\ell=0}^k D_\ell^m(x), \end{aligned} \quad (60)$$

where

$$D_\ell^m(x) \equiv \binom{m}{\ell} x^{m-\ell} (1-x)^\ell. \quad (61)$$

Equation (61) represents a cumulative sum, each term of which can be computed recursively. For example, when x is bounded away from zero,

$$D_\ell^m(x) = \left(\frac{m+1}{\ell} - 1 \right) \cdot \left(\frac{1}{x} - 1 \right) D_{\ell-1}^m(x), \quad (62)$$

for $\ell = 1, 2, \dots$, with $D_0^m(x) = x^m$. A similar recursion holds for small x , computed backwards from $D_m^m(x) = (1-x)^m$.

An explicit representation of the polynomials $P_k^m(x)$ is given by

$$P_k^m(x) = 1 - (1-x)^{k+1} \sum_{\ell=0}^{m-k-1} \binom{k+\ell}{\ell} x^\ell. \quad (63)$$

It may be observed that $P_k^m(x) = 1$ whenever $m \leq k$, and that $P_k^m(x) \geq 0$ for all $0 \leq x \leq 1$. These polynomials also satisfy the recurrence relation

$$P_k^{m+1}(x) = P_k^m(x) + \frac{m}{k} (1-x) [P_{k-1}^m(x) - P_{k-1}^{m-1}(x)] \quad (64)$$

for $k, m = 1, 2, \dots$, subject to

$$P_0^m(x) = x^m, \quad P_k^0(x) = 1, \quad P_k^1(x) = 1 - (1-x)\delta_{k0}. \quad (65)$$

5.3. Evaluation In order to achieve good numerical behaviour as $\nu \rightarrow 1$, it is convenient to work with the scaled integrals $\hat{R}_{n+1}^m \equiv (-r_{\text{lo}})^n R_{n+1}^m$, for which we have the well-behaved series representation

$$\begin{aligned} \hat{R}_{n+1}^m &= \frac{1}{(z_2 - z_1)(1 - z_1/z_0)} \cdot \left(\frac{r_{\text{lo}}}{(z_2 - z_1)^2} \right)^n \\ &\quad \times \sum_{k=0}^n \binom{2n-k}{n} \left(\frac{z_2 - z_1}{1 - z_1/z_0} \right)^k z_0^{m-k-1} P_k^m(z_1/z_0). \end{aligned} \quad (66)$$

Thus, we consider the computation of the vectors

$$\hat{\mathbf{R}}^{(m)} \equiv [\hat{R}_1^m, \hat{R}_2^m, \dots, \hat{R}_{N+1}^m]^\top. \quad (67)$$

To assist with this, we define the constant

$$\kappa \equiv \frac{1}{(z_2 - z_1)(1 - z_1/z_0)}, \quad (68)$$

the diagonal matrices

$$\begin{aligned} A &\equiv \text{diag}[a^0, a^1, \dots, a^N], & a &\equiv r_{\text{lo}}/(z_2 - z_1)^2, \\ B &\equiv \text{diag}[b^0, b^1, \dots, b^N], & b &\equiv (z_2 - z_1)/(1 - z_1/z_0), \end{aligned} \quad (69)$$

and the combinatorial matrix

$$C_{nk} \equiv \binom{2n-k}{n} \quad (70)$$

provided $k \leq n$ and is zero otherwise. We also introduce the polynomial vectors

$$\mathbf{P}^{(m)} \equiv [P_0^{(m)}, P_1^{(m)}, \dots, P_N^{(m)}]^\top, \quad P_k^{(m)} \equiv z_0^{m-k-1} P_k^m(z_1/z_0). \quad (71)$$

Then, we can write (66) as

$$\begin{aligned} \hat{\mathbf{R}}^{(m)} &= \kappa \cdot ACBP^{(m)} \\ &= \kappa \cdot (ACA^{-1}) \cdot AB \cdot \mathbf{P}^{(m)}. \end{aligned} \quad (72)$$

At this point, we note that the product AB is the diagonal matrix of increasing powers

$$AB = \text{diag}[\gamma^0, \gamma^1, \dots, \gamma^N], \quad \gamma \equiv r_{\text{lo}}/[(z_2 - z_1)(1 - z_1/z_0)], \quad (73)$$

and that $(ACA^{-1})_{nk} = a^{n-k} C_{nk}$, which is easily computed by observing the cumulative product form

$$a^\ell \binom{n+\ell}{n} = \prod_{j=1}^{\ell} (1 + n/j) a. \quad (74)$$

If we combine the column vectors $\hat{\mathbf{R}}^{(m)}$ and $\mathbf{P}^{(m)}$ into respective matrices, so that

$$\begin{aligned} \hat{\mathbf{R}} &\equiv [\hat{\mathbf{R}}^{(0)}, \hat{\mathbf{R}}^{(1)}, \dots, \hat{\mathbf{R}}^{(M)}], \\ \mathbf{P} &\equiv [\mathbf{P}^{(0)}, \mathbf{P}^{(1)}, \dots, \mathbf{P}^{(M)}], \end{aligned} \quad (75)$$

then we obtain the matrix equation

$$\hat{\mathbf{R}} = \kappa \cdot (ACA^{-1}) \cdot AB \cdot \mathbf{P}. \quad (76)$$

In Figure 2, we plot the queue-length PMF for the low-priority arrivals, as the negative base-10 logarithm, for total traffic intensity $r = 0.99$ and a range of hifrac values ν . Overlaid, are the asymptotic curves in the large queue-length limit. This is given by

$$f_{\text{lo}}(n) \underset{n \rightarrow \infty}{\sim} \sqrt{\frac{1-r}{\pi r}} \cdot \frac{r^n}{\sqrt{n}}, \quad (77)$$

when $r_{\text{hi}} = r^2$ (or equivalently $\nu = r$). Otherwise, the low-priority marginal PMF can be decomposed into two components according to

$$f_{\text{lo}}(n) = f_{\text{pol}}(n) \cdot \Theta(r^2 - r_{\text{hi}}) + f_{\text{cut}}(n), \quad (78)$$

where $\Theta(x)$ denotes the Heaviside function such that $\Theta(x) = 1$ for $x \geq 0$ and vanishes otherwise. The large- n behaviour of these components is given by

$$\begin{aligned} f_{\text{pol}}(n) &\underset{n \rightarrow \infty}{\sim} \left[1 - \frac{r(1-r)}{r_{\text{lo}}} \right] (1-r) r^{n-1}, \\ f_{\text{cut}}(n) &\underset{n \rightarrow \infty}{\sim} \frac{(\sqrt{r_{\text{hi}}/r_{\text{lo}}})^{1/2}}{2\sqrt{\pi r}} \cdot \frac{1-r}{(\chi - 1/r) \chi^{n-1/2} n^{3/2}}, \end{aligned} \quad (79)$$

where $\chi \equiv 1 + (1 - \sqrt{r_{\text{hi}}})^2 / r_{\text{lo}} > 1/r$. The derivation of these results, which will be presented in a forthcoming paper, follows directly from the pole/cut integral representation of the distribution, mentioned in Section 4. The computed points, represented by the coloured dots, are interpolated by black curves. The asymptotic curves are indicated by a coloured dashed line-style. Thus, when the interpolation between the data points becomes coloured, this indicates that the agreement between the computation and asymptotic limit is within the linewidth of the graph. In Figure 3, we plot the queue-length PMF for the low-priority arrivals, as the negative base-10 logarithm, for the case of total traffic intensity $r = 0.99$ and fraction of high-priority arrivals $\nu = 0.95$, where asymptotic behaviour is slow to set in. We see that the computation remains robust up to a queue length of at least $n = 1000$ which lies deep in the asymptotic region. In Figure 4, we plot a two-dimensional map of the joint probability distribution $f(n, m)$ of the queue lengths, for total traffic intensity $r = 0.75$ and fraction of high-priority arrivals $\nu = 0.9$. A logarithmic scaling has been applied, such that $f(n, m) \leftarrow \max\{0, 1 + \log_{10}(f(n, m)/f_{\text{max}})/20\}$, where $f_{\text{max}} \equiv \max\{f(n, m)\}$.

5.4. Limiting Cases The $\nu \rightarrow 0$ limiting behaviour of the R-integrals is given by

$$R_n^m \underset{\nu \rightarrow 0^+}{\sim} \begin{cases} 0 & \text{for } m > n \\ (-1)^{n-1} [1 - r^{n-1}/(1+r)^{2n-1}] & \text{for } m = n \\ (-1)^{n-1} / \nu^{n-m} & \text{for } m < n \end{cases} . \quad (80)$$

At the opposite extreme, for $\nu = 1$, we have $z_0 = 1$, $z_1 = r$, $z_2 = 1$, in which case

$$\frac{z_2 - z_1}{1 - z_1/z_0} = 1 . \quad (81)$$

It follows that

$$R_{n+1}^m \underset{\nu \rightarrow 1^-}{\sim} \frac{(-1)^n}{(1-r)^{2n+2}} \sum_{k=0}^n \binom{2n-k}{n} P_k^m(r) . \quad (82)$$

Equation (80) shows that the R-integrals become singular for small ν when $m < n$. This is one reason for the numerical instability of the recurrence relations (51), given that the seed values always reside in this region.

6. Numerical Tests Various tests can be applied to quantify the numerical performance of the algorithm for the computation of the joint PMF.

6.1. Aggregation Test The aggregated queue-length distribution describes the total number of entities in the queue, regardless of priority level. This is equivalent to the queue-length distribution of the basic M/M/c queueing model with traffic intensity $r = r_{\text{lo}} + r_{\text{hi}}$, which is known to be a simple geometric distribution. Hence, the exact aggregate PMF is given by

$$f_{\text{agg}}^{(\text{ex})}(k) = (1-r)r^k , \quad (83)$$

for $k = 0, 1, 2, \dots$

One diagnostic test of the R-integral computational methodology is to check how well the aggregate PMF constructed from the computed joint PMF reproduces the exact result. This test is more convenient than similarly testing against the marginals as only a finite summation is required. Considering the joint PMF as a matrix whose rows and columns are labelled by its integer arguments, values of the aggregate PMF are given by successive finite sums along the anti-diagonals. Specifically, in terms of the R-integrals, the aggregate PMF is expressed as

$$\begin{aligned} f_{\text{agg}}(k) &= \sum_{n=0}^k f(n, k-n) \\ &= \frac{1-r}{r} \sum_{n=0}^k (-r_{\text{lo}})^n (\Delta R_{n+1}^{k+2} - r_{\text{hi}} \Delta R_{n+1}^k) , \end{aligned} \quad (84)$$

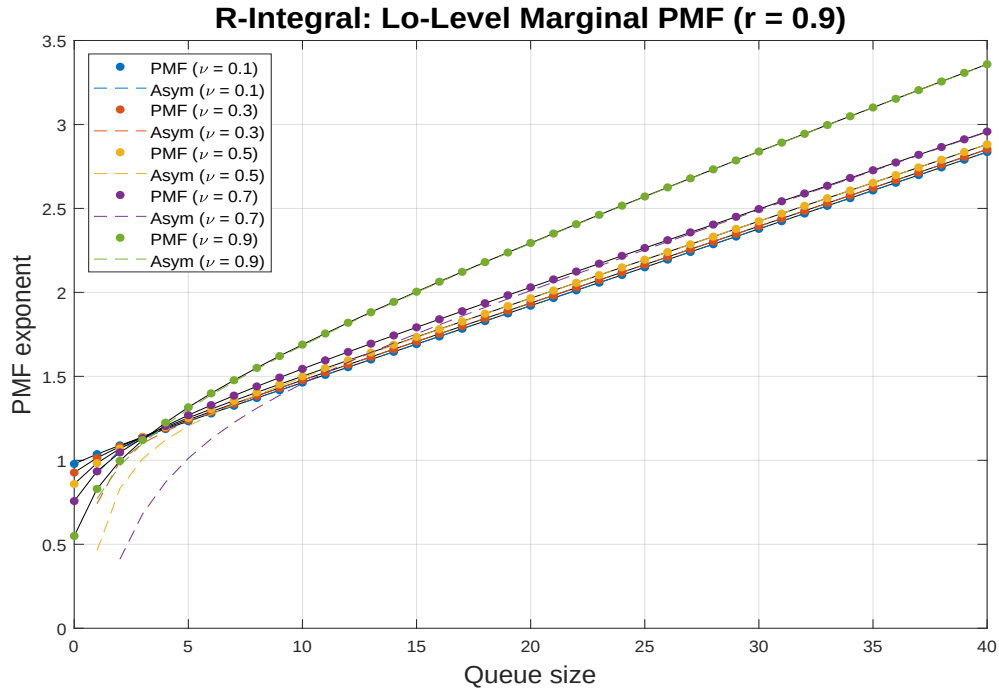


FIGURE 2.

Note. Queue-length PMF for the low-priority arrivals, plotted as the negative base-10 logarithm, for total traffic intensity $r = 0.9$ and a range of hifrac values (ν). Asymptotic curves for the large queue-length limit are overlaid.

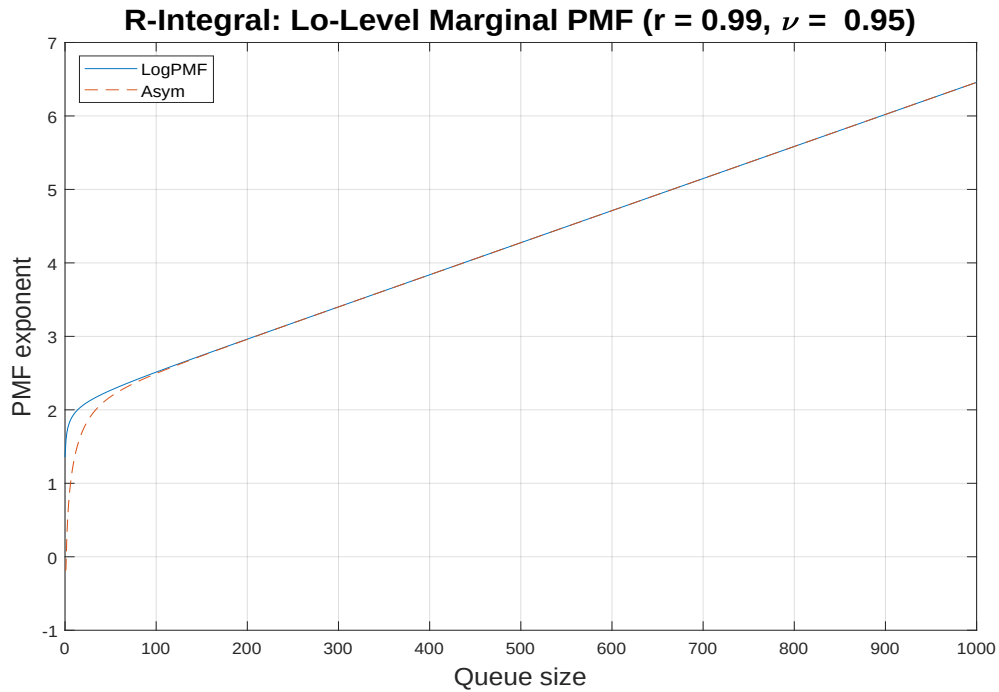


FIGURE 3.

Note. Queue-length PMF for the low-priority arrivals, plotted as the negative base-10 logarithm, for total traffic intensity $r = 0.99$ and fraction of high-priority arrivals $\nu = 0.9$, with queue lengths extending far into the asymptotic region. It is compared with the exact asymptotic curve in the large queue-length limit.

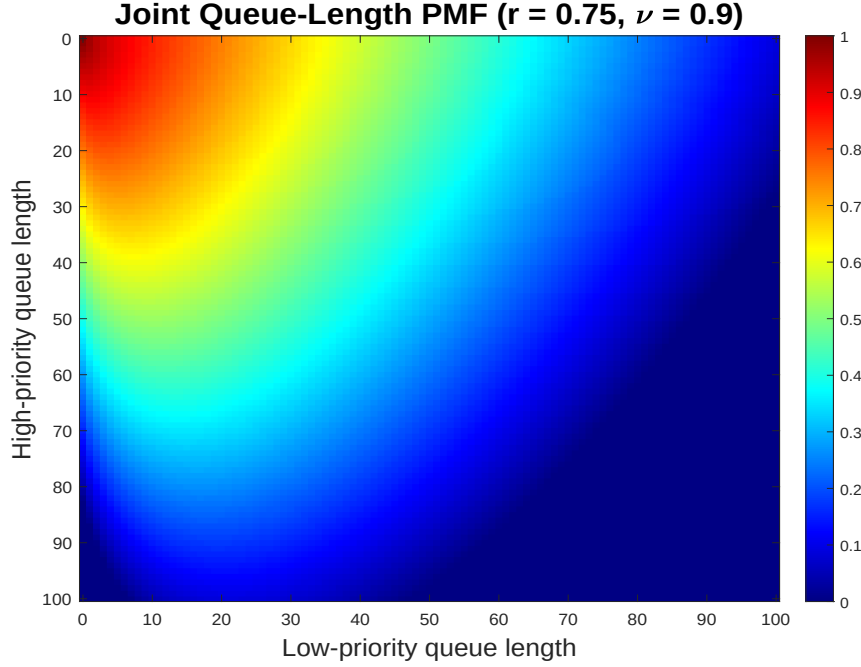


FIGURE 4.

Note. 2D map of the joint probability distribution of the queue lengths, with a logarithmic scaling applied, for total traffic intensity $r = 0.75$ and fraction of high-priority arrivals $\nu = 0.9$.

for $k = 0, 1, 2, \dots$

We then consider the measure of performance (MOP)

$$\Xi_{\text{agg}} \equiv -\max_{k \geq 0} \left\{ \log_{10} (|\ln(f_{\text{agg}}(k)) - \ln(f_{\text{agg}}^{(\text{ex})}(k))|) \right\}, \quad (85)$$

where the maximum is taken over all values $0 \leq k \leq n_{\text{lim}}$ such that $f_{\text{agg}}^{(\text{ex})}(k) > p_{\text{lim}} > 0$. Since we are working in double-precision arithmetic⁴, all MOPs of this kind are capped at a maximum allowed value of 16. The interpretation of Ξ_{agg} (and similarly for all of the subsequent MOPs) is that it indicates the number of decimal places of numerical agreement in the worst case.

6.2. Xhi-Test The exclusively-high distribution, defined by $f_{\text{xhi}}(m) \equiv f(0, m)$, gives the probability of finding m high-priority clients in the queue and no low-priority clients. An exact expression for the exclusively-high probability is given by

$$f_{\text{xhi}}^{(\text{ex})}(m) = (1 - r)(r_{\text{hi}}/z_2)^m, \quad (86)$$

for $m = 0, 1, 2, \dots$. It is simple to calculate directly as the R-integral has only a simple pole when $n = 0$. One should note that $f_{\text{xhi}}(m)$ is not a proper PMF since $\sum_{m=0}^{\infty} f_{\text{xhi}}(m) < 1$, unless $\nu = 1$, but can be turned into a conditional PMF by means of an overall scale factor.

In terms of the R-integrals, the exclusively-high PMF is expressed as

$$f_{\text{xhi}}(m) = \frac{1 - r}{r} (\Delta R_1^{m+2} - z_1 z_2 \Delta R_1^m), \quad (87)$$

⁴ All computation is performed in MATLAB R2020a, which implements IEEE Standard 754 for double precision.

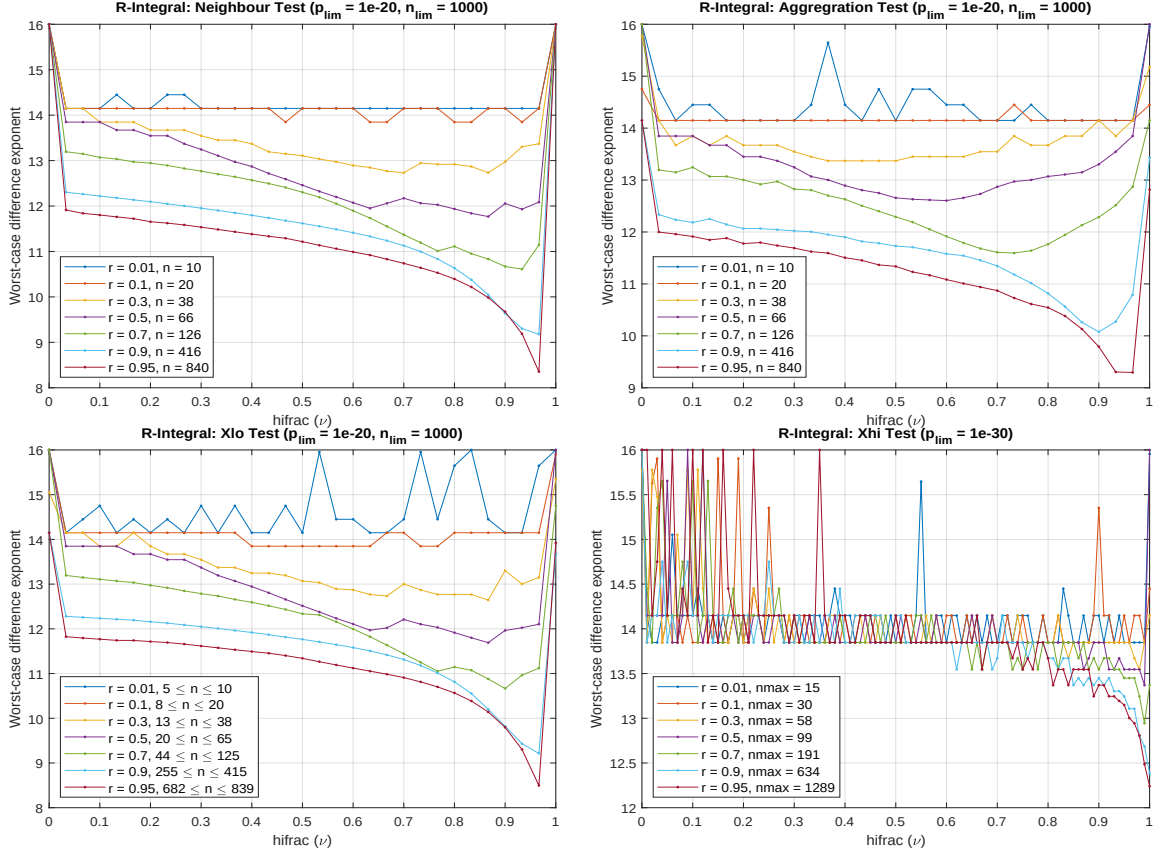


FIGURE 5.

Note. Four tests of the joint probability distribution of the queue lengths as functions of the fraction of high-priority arrivals ν , across a wide range of values for the total traffic intensity r , as displayed. The values on the vertical axes indicate the number of decimal places of agreement.

and we consider the MOP

$$\Xi_{xhi} \equiv -\max_{m \geq 0} \left\{ \log_{10} \left(\left| \ln(f_{xhi}(m)) - \ln(f_{xhi}^{(ex)}(m)) \right| \right) \right\}, \quad (88)$$

where the maximum is taken over all values $0 \leq m \leq n_{lim}$ such that $f_{xhi}^{(ex)}(m) > p_{lim} > 0$.

6.3. Xlo-Test Checking whether the computed joint PMF gives rise to the correct marginal distribution, numerically, is not a convenient enterprise as it necessitates an infinite summation. However, it is possible to devise an alternative test that checks the consistency of the numerical low-priority marginal with the numerically computed joint PMF. In the xlo-test, we relate the exclusively-low distribution $f_{xlo}(n)$ with the low priority marginal $f_{lo}(n)$. To achieve this, we consider the PGF (8) recast into the form

$$g_m(p) = (1-r) \frac{1-\lambda_1(p)}{\lambda_2(p)-r} \cdot \lambda_1^m(p). \quad (89)$$

Specialized to the case $m=0$, this may be expressed as

$$g_0(p) = (1-r) \left[1 + r_{lo} p \cdot \frac{1}{\lambda_2(p)-r} \right]. \quad (90)$$

Since the PGF of the low-priority marginal is given by

$$g_{\text{lo}}(p) = \sum_{m=0}^{\infty} g_m(p) = \frac{1-r}{\lambda_2(p)-r}, \quad (91)$$

we arrive at the result

$$g_0(p) = 1 - r + r_{\text{lo}} p \cdot g_{\text{lo}}(p). \quad (92)$$

There is a generalization of this result to non-zero values of m that relates $g_m(p)$ to $g_{\text{lo}}(p)$. Its derivation is presented in the Appendix. From the relationships

$$g_0(p) = \sum_{n=0}^{\infty} p^n f_{\text{xlo}}(n), \quad g_{\text{lo}}(p) = \sum_{n=0}^{\infty} p^n f_{\text{lo}}(n), \quad (93)$$

we can equate powers to read off that

$$f_{\text{xlo}}(0) = 1 - r, \quad f_{\text{xlo}}(n) = r_{\text{lo}} \cdot f_{\text{lo}}(n-1), \quad (94)$$

for $n = 1, 2, \dots$, or, equivalently,

$$f_{\text{xlo}}(n) = (1-r)\delta_{n0} + (1-\delta_{n0})r_{\text{lo}} \cdot f_{\text{lo}}(n-1), \quad (95)$$

for $n = 0, 1, 2, \dots$, where we can formally set $f_{\text{lo}}(-1) \equiv 0$. One should note that $f_{\text{xlo}}(n)$ is not a proper PMF since

$$\sum_{n=0}^{\infty} f_{\text{xlo}}(n) = 1 - r_{\text{hi}}, \quad (96)$$

but can be turned into a conditional PMF by means of an overall scale factor.

For the xlo-test, the LHS of (95) is taken to be given by (50) and is compared with the RHS of (95) where the marginal PMF $f_{\text{lo}}(n)$ is expressed in terms of the R-integrals via (49). The relevant MOP is taken to be

$$\Xi_{\text{xlo}} \equiv -\max_{n>0} \{ \log_{10} (| \ln(f_{\text{xlo}}(n)) - \ln(r_{\text{lo}} f_{\text{lo}}(n-1)) |) \}, \quad (97)$$

where the maximum is taken over all values $0 < n \leq n_{\text{lim}}$ such that $f_{\text{xlo}}(m) > p_{\text{lim}} > 0$.

6.4. Nearest-Neighbour Test A direct consequence of the recurrence relations for the R-integrals is that the joint PMF at any given interior point (n, m) is a positively weighted sum of the joint PMF values at three of its four nearest neighbours:

$$f(n, m) = \frac{1}{1+r} [f(n, m+1) + r_{\text{lo}} f(n-1, m) + r_{\text{hi}} f(n, m-1)], \quad (98)$$

for all $m, n > 0$.

In order to apply the neighbour test, we first compute the joint PMF $f(n, m)$ on a 2D grid of points (n, m) from (48). Next, we use these values to compute the RHS of (98), which we shall denote $f_{\text{nn}}(n, m)$. Then, we consider the MOP

$$\Xi_{\text{nn}} \equiv -\max_{m, n > 0} \{ \log_{10} (| \ln(f(n, m)) - \ln(f_{\text{nn}}(n, m)) |) \}, \quad (99)$$

where the maximum is taken over all values $0 < m, n \leq n_{\text{lim}}$ such that $f(n, m) > p_{\text{lim}} > 0$.

6.5. Quadratic Test In this test, we compare the results for the joint queue-length PMF computed from the R-integral (denoted $f_{ri}(n, m)$) with that computed by the quadratic recurrence (denoted $f_{qr}(n, m)$). The MOP is taken to be the number of decimal places of agreement, as given by

$$\Xi_{qr} \equiv - \max_{m, n > 0} \{ \log_{10} (| \ln(f_{ri}(n, m)) - \ln(f_{qr}(n, m)) |) \} , \quad (100)$$

where the maximum is taken over all values $0 < m, n \leq n_{lim}$ such that $f(n, m) > p_{lim} > 0$.

6.6. Results Figure 5 presents the results of the numerical tests. The MOP values relevant to the R-integral computations are displayed on the vertical axis against the full range of high-priority arrival fraction (hifrac) ν on the horizontal axis. Individual curves are plotted for a discrete collection of traffic intensities, spanning a wide range. Agreement always exceeds eight decimal places, and is generally much higher. The nearest-neighbour and xlo-tests check the internal consistency of the computations, while the aggregation and xhi-tests check against exact analytical results.

The maximum queue occupancy to be examined was taken to be $n_{lim} = 1000$. PMF intervals examined included everything down to a tail value of $p_{lim} = 10^{-20}$ except in the xhi-test where $p_{lim} = 10^{-30}$ was used.

Figure 6 presents the results of comparing the joint queue-length distribution computed from the R-integral with that computed by the quadratic recurrence. The close agreement observed implies a high level of accuracy for each method across the complete range of parameters. Worst case accuracy occurs when both the traffic intensity r and hifrac ν approach unity. In Table 1, we present results that investigate this region in more detail. Values of r close to unity have been reported to be problematic for the matrix-geometric approach [17, 18]. The table shows that both of the present methods behave well in this region. The fourth and fifth columns indicate the smallest rectangular subset $[0, n_{lo}] \times [0, n_{hi}]$ of $[0, n_{lim}] \times [0, n_{lim}]$ that contains all grid points (m, n) with probability greater than $p_{lim} = 10^{-20}$. A value of $n_{lim} = 1000$ in one or both columns indicates that p_{lim} was not attained in some direction. The last column is the minimum probability that was achieved over all considered grid points whose probability values exceed p_{lim} . Computation time for the quadratic recurrence method is two orders of magnitude faster than for the R-integral method. Finally, Figure 7 repeats the quadratic test as described above, but for the low-priority marginal PMFs, with the distribution arising from the quadratic recurrence computed by the algorithm of (23). The legend indicates the of range maximum queue lengths n that had to be considered across the full range of hifrac values ν in order to attain the limiting probability level $p_{lim} = 10^{-20}$ for the given traffic intensity r . Agreement between the R-integral and quadratic recurrence approaches is observed to exceed ten decimal places in the worst case..

The exact results for the queue-lengths distributions derived here were also tested against Monte-Carlo simulation. Excellent agreement was found across the entire parametric domain. Details will be presented elsewhere. We have also checked against the results in Table 3 of [8] (where the service times are equal) to find complete agreement. There, in the case of the present problem, the quantity P_Q is related to the no-wait probability P_{NW} given in (3) by $P_Q = 1 - P_{NW}$, and $p(0, 0)$ is the probability that the system is empty, given by

$$\frac{1}{p(0, 0)} = \frac{(rN)^N}{N!} \left[\frac{1}{1-r} + \Gamma_{scl}(rN, N) \right] , \quad (101)$$

where

$$\Gamma_{scl}(x, \nu) \equiv \frac{\nu e^x}{x^\nu} \int_x^\infty dt \, t^{\nu-1} e^{-t} \quad (102)$$

is the scaled upper incomplete gamma function as implemented in MATLAB. In the present problem, neither of these quantities depend on the priority structure. We relate the mean waiting times given in the table to the mean queue lengths via Little's law.

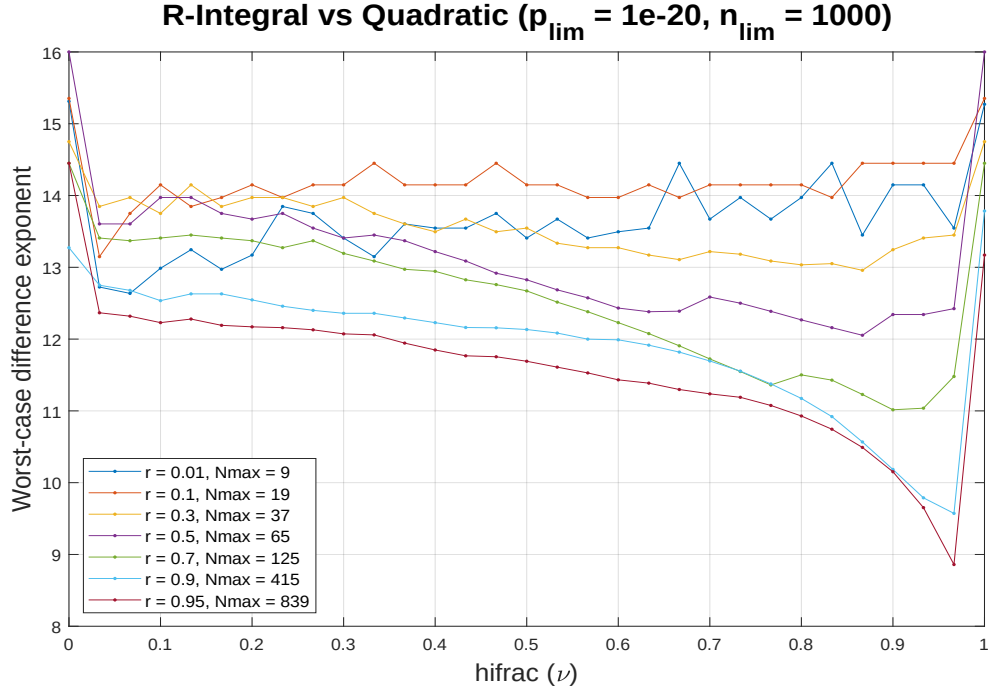


FIGURE 6.

Note. Comparison of the joint probability distribution of the queue lengths as computed from the R-integral and from the quadratic recurrence. The MOP is plotted as a function of the fraction of high-priority arrivals ν , across a wide range of values for the total traffic intensity r , as displayed. The values on the vertical axes indicate the number of decimal places of agreement.

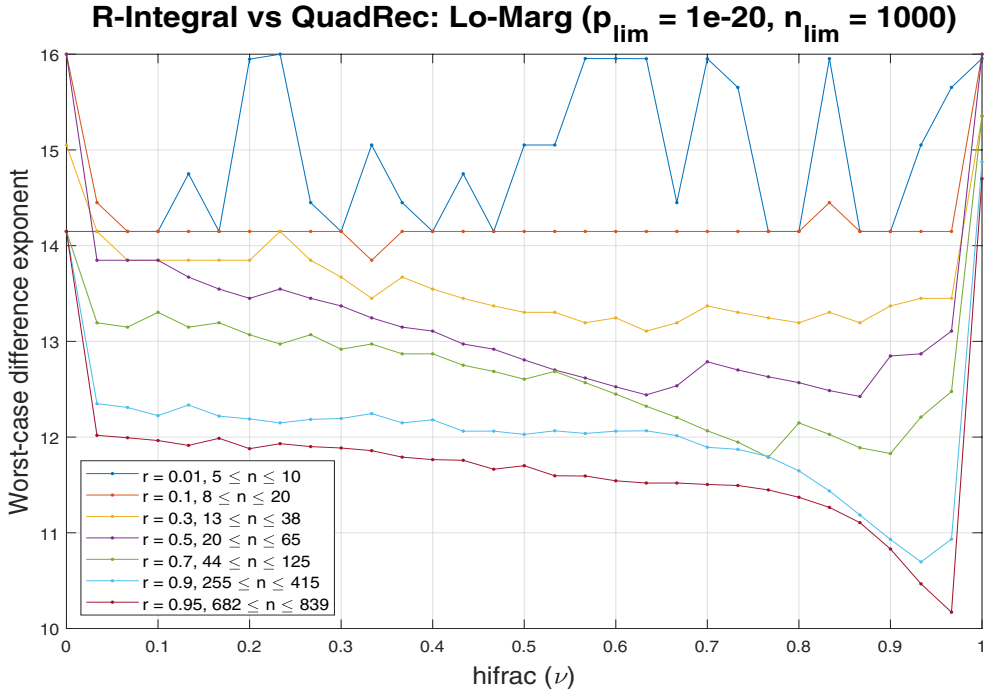


FIGURE 7.

Note. Comparison of the low-priority marginal distribution of the queue lengths as computed from the R-integral and from the quadratic recurrence. The MOP is plotted as a function of the fraction of high-priority arrivals ν , across a wide range of values for the total traffic intensity r , as displayed. The values on the vertical axes indicate the number of decimal places of agreement.

TABLE 1. Joint PMF Comparison

r	ν	Ξ_{qr}	n_{hi}	n_{lo}	p_{min}
0.99	0.95	9.3279	609	1000	1.0000×10^{-20}
	0.99	8.1611	1000	1000	1.0000×10^{-20}
	0.999	6.6633	1000	1000	1.0000×10^{-20}
	1.00	11.7428	1000	0	4.3171×10^{-7}
0.999	0.95	9.4247	685	1000	1.0000×10^{-20}
	0.99	8.4169	1000	1000	1.0017×10^{-20}
	0.999	7.2251	1000	1000	6.6926×10^{-18}
	1.00	9.6972	1000	0	3.6770×10^{-4}
0.9999	0.95	9.4344	657	1000	1.0000×10^{-20}
	0.99	8.4361	1000	1000	1.0000×10^{-20}
	0.999	7.2455	1000	1000	1.0540×10^{-18}
	1.00	7.8504	1000	0	9.0483×10^{-5}

7. Conclusions Simple methods for highly accurate computation of the joint and marginal distributions for a non-preemptive two-level priority queue have been developed. Explicit closed-form representations for the joint and marginal PMFs have also been derived, something that has not been achieved previously. Future work could entail extension of the present methods to unequal services rates among the priority levels.

Appendix. Convolutional Form

In this appendix, we derive a relationship between joint PMF $f(n, m)$ and the low-priority marginal $f_{\text{lo}}(n)$. One may observe the general structure

$$g_m(p) = A^{(m)}(p) + B^{(m)}(p) \cdot g_{\text{lo}}(p), \quad (103)$$

for polynomials

$$A^{(m)}(p) = \sum_{n=0}^m A_n^{(m)} p^n, \quad B^{(m)}(p) = \sum_{n=0}^m B_n^{(m)} p^n. \quad (104)$$

Thus, we obtain the convolutional form for the joint PMF:

$$f(n, m) = A_n^{(m)} + \sum_{k=0}^n B_k^{(m)} \cdot f_{\text{lo}}(n - k), \quad (105)$$

which generalizes the relationship between f_{xlo} and f_{lo} given in (95) to non-zero values of m . One may also note the special case

$$f_{\text{xhi}}(m) \equiv f(0, m) = A_0^{(m)} + B_0^{(m)} f_{\text{lo}}(0). \quad (106)$$

In what follows, we derive explicit expressions for $A^{(m)}(p)$ and $B^{(m)}(p)$.

Since $\lambda_1(p)$ satisfies a quadratic equation, we have that $\lambda_1^m(p) = \alpha_m(p)\lambda_1(p) + \beta_m(p)$ for some polynomials $\alpha_m(p)$, $\beta_m(p)$. The fact that α_m, β_m are polynomials follows from examining λ_1^2 . Let us now recall that the Chebyshev polynomials of the first and second kind, $T_n(x)$ and $U_n(x)$ respectively, may be expressed as

$$\begin{aligned} T_n(x) &= \frac{1}{2} \left[(x + \sqrt{x^2 - 1})^n + (x - \sqrt{x^2 - 1})^n \right], \\ \sqrt{x^2 - 1} U_{n-1}(x) &= \frac{1}{2} \left[(x + \sqrt{x^2 - 1})^n - (x - \sqrt{x^2 - 1})^n \right], \end{aligned} \quad (107)$$

which implies the identity

$$(x - \sqrt{x^2 - 1})^n = T_n(x) - xU_{n-1}(x) + (x - \sqrt{x^2 - 1}) \cdot U_{n-1}(x). \quad (108)$$

Since we have

$$\lambda_1(p) = b(p) - \sqrt{b^2(p) - r_1}, \quad b(p) \equiv (1 + r - r_2 p)/2, \quad (109)$$

it follows that

$$\frac{\lambda_1^m(p)}{r_1^{m/2}} = T_m(x(p)) - x(p) \cdot U_{m-1}(x(p)) + \frac{\lambda_1(p)}{\sqrt{r_1}} \cdot U_{m-1}(x(p)), \quad (110)$$

with $x(p) \equiv b(p)/\sqrt{r_1}$. Consequently,

$$\begin{aligned} \alpha_m(p) &= r_1^{(m-1)/2} U_{m-1}(x(p)), \\ \beta_m(p) &= r_1^{m/2} [T_m(x(p)) - x(p) \cdot U_{m-1}(x(p))], \end{aligned} \quad (111)$$

for $x = 0, 1, \dots$, with $U_{-1}(x) \equiv 0$. The first few α -coefficients are given by

$$\alpha_0(p) = 0, \quad \alpha_1(p) = 1, \quad \alpha_2(p) = 1 + r - r_2 p. \quad (112)$$

The first few β -coefficients are given by

$$\beta_0(p) = 1, \quad \beta_1(p) = 0, \quad \beta_2(p) = -r_1. \quad (113)$$

Using identities satisfied by the Chebyshev polynomials, (111) can be simplified as

$$\alpha_m(p) = r_1^{(m-1)/2} U_{m-1}(x(p)), \quad \beta_m(p) = -r_1^{m/2} U_{m-2}(x(p)), \quad (114)$$

which implies the relationship $\beta_m(p) = -r_1 \alpha_{m-1}(p)$, where we formally set $\alpha_{-1}(p) \equiv 1$.

Noting that

$$\begin{aligned} (1 - \lambda_1) \lambda_1^m &= (\alpha_m - \alpha_{m+1}) \lambda_1 + (\beta_m - \beta_{m+1}) \\ &= -(\alpha_m - \alpha_{m+1})(\lambda_2 - r) + (1 - r_2 p)(\alpha_m - \alpha_{m+1}) + (\beta_m - \beta_{m+1}), \end{aligned} \quad (115)$$

followed by substitution into the representation

$$g_m(p) = \frac{1-r}{\lambda_2-r} \cdot (1 - \lambda_1) \lambda_1^m, \quad (116)$$

yields

$$g_m(p) = -(1-r)(\alpha_m - \alpha_{m+1}) + g_{10}(p) [(1 - r_2 p)(\alpha_m - \alpha_{m+1}) + (\beta_m - \beta_{m+1})], \quad (117)$$

from which we can read off

$$\begin{aligned} A^{(m)}(p) &= (1-r) [\alpha_{m+1}(p) - \alpha_m(p)], \\ B^{(m)}(p) &= (1-r_2 p) [\alpha_m(p) - \alpha_{m+1}(p)] + [\beta_m(p) - \beta_{m+1}(p)]. \end{aligned} \quad (118)$$

In terms of Chebyshev polynomials, this becomes

$$\begin{aligned} A^{(m)}(p) &= -(1-r) r_1^{(m-1)/2} [U_{m-1}(x(p)) - \sqrt{r_1} U_m(x(p))], \\ B^{(m)}(p) &= -\frac{1-r_2 p}{1-r} \cdot A^{(m)}(p) + \frac{r_1}{1-r} \cdot A^{(m-1)}(p), \end{aligned} \quad (119)$$

for $m = 0, 1, \dots$, where we set $U_{-1}(x) \equiv 0$, $U_{-2}(x) \equiv -1$. As a sanity check, it is straightforward to confirm that $A^{(m)}(1) + B^{(m)}(1) = (1-r_1)r_1^m$. Another check is given by

$$\begin{aligned} \sum_{m=0}^{\infty} A^{(m)}(p) &= \begin{cases} (1-r)/(1-r_1) & \text{for } p = 1 \\ 0 & \text{for } p \neq 1 \end{cases}, \\ \sum_{m=0}^{\infty} B^{(m)}(p) &= \begin{cases} (r-r_1)/(1-r_1) & \text{for } p = 1 \\ 1 & \text{for } p \neq 1 \end{cases}. \end{aligned} \quad (120)$$

Acknowledgments. The authors gratefully acknowledge useful discussions with Dr. Stephen Bocquet.

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